Recap

1. What is the output of the following code?

```
fruit = 'apple'
if fruit == 'banana':
    print(5%2)
elif fruit == 'avocado':
    print(5//2)
else:
    print(5**2)
```

you can write the code below and check your answer

```
#test your code here
fruit = 'apple'
if fruit == 'banana': #false
    print(5%2)
elif fruit == 'avocado': #false
    print(5//2)
else:
    print(5**2)

25
```

1. what is the output of the given code?

```
x = 0
if True:
    x = 1
else:
    x = 2
print(x)

x = 0
if True:
    x = 1
else:
    x = 2
print(x)

1
```

1. Trace the following code and guess what the output is

```
nums = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
for val in nums:
    if val % 2 == 0:
      print(val)
```

check your answer in the next cell

```
#test the code here
nums = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
for val in nums: # val = 1, 2, 3...10
    if val % 2 == 0: # 1%2 = 1, 2%2 = 0
      print(val) #2, 4, 6, 8, 10
```

```
2
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1. We use for loop when we are not sure how many times the loop is iterated(True/False)

```
False
#for loop: we don't usually use it when we don't know how many
iterations
#while loop: can be used even if we don't know how many times it is
iterated
```

1. Trace the following code and find what the value of sum is at the end of the loop

```
sum = 0

values = [3, 18, 2, -2, 9]

for i in range(3):
  sum += values[i]
print(sum)

sum = 0

values = [3, 18, 2, -2, 9]

for i in range(3): # 0, 1, 2
  sum += values[i] # sum += values[2], sum +=2, sum = 23
print(sum)
```

```
23
```

1. given the above hint, try to write a code to find the sum of all elements in a list.

```
numbers = [28, 89, 100, 90, 23, 25, 18]

#our list name is numbers
numbers = [28, 89, 100, 90, 23, 25, 18]

#write your code below
```

1. change the code above to while loop if you used for loop or vice versa:

```
#write your code here
```

While Loops

While loop does the same thing as for loop!

- **The main difference is, they are often used when we do not know how many times the iteration will happen.**
- While loops repeat as long as a certain `boolean condition is met`.

syntax

```
while CONDITION:
  Do Something

#Example:
count = 0
while count < 5: #the loop continues as long as our variable count is less than 5
    print(count)
    count += 1  # This is the same as count = count + 1

while condition(True/False):
  if the condition is true, it keeps looping.

name = "Samson"
count = 0
while count < 5: #count = 0, 0 < 5 = True
  print(name) #samson.. prints samson forever.. infinite loop
```

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Streaming output truncated to the last 5000 lines.
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---------------------------------------------------------------------------
KeyboardInterrupt                         Traceback (most recent call last)
/tmp/ipython-input-1828/405865097.py in <cell line: 0>()
      2 count = 0
      3 while count < 5: #count = 0, 0 < 5 = True
----> 4   print(name) #samson

/usr/local/lib/python3.12/dist-packages/ipykernel/iostream.py in write(self, string)
    548         else:
    549
--> 550             is_child = not self._is_master_process()
    551             # only touch the buffer in the IO thread to avoid races
    552             with self._buffer_lock:

/usr/local/lib/python3.12/dist-packages/ipykernel/iostream.py in _is_master_process(self)
    436
    437     def _is_master_process(self):
--> 438         return os.getpid() == self._master_pid
    439
    440     def set_parent(self, parent):
```

```
KeyboardInterrupt:
```

```
name = "Samson"
count = 0
while count < 5: #count = 0, 1 < 5 = True... 5<5 = False.. stops
  print(name) #Samson Samson Samson Samson Samson
  count += 1 #count = 5
```

```
Samson
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```

```
count = 0
while count < 5: # 0<5:True, 1 < 5:True ... 5 < 5: False
    print(count) # 0 1 2 3 4
     count += 1 #count = 5
```

```
0
1
2
3
4
```

Caution: when using while loop, you should be very careful about stopping condition!

`If the stopping condition is never met, the loop goes forever!!`

```
'''
counter = 2
while counter > 0: 2>0
  print(i)
'''
#in the above loop counter is never updated and it's always 2, which
means the condition counter > 0 is alwasy true! so the loop goes
forever!
```

```
{"type":"string"}
```

```
# Print i as long as i is less than 6:
i = 1
while i < 6:
  print(i)
  i += 1 #increment(updating i so that it goes to the stopping stage)
```

```
1
2
3
4
5
```

```
fruits = ['orange', 'banana', 'apple', 'pineapple']
#printing all fruits
y = 0
while y < len(fruits):
    print(fruits[y])
    y += 1
```

```
orange
banana
apple
pineapple
```

Break and Continue

- break: as it's name suggests stop or breaks while or for loop
- continue: will skip the current block and return to the 'while' or 'for' loop statement

```
# Prints out 0,1,2,3,4
count = 0
while True:
    print(count) #0 1 2 3 4
    count += 1 #count = 5
    if count >= 5: #5 >= 5: True
        break #stop and leave the loop
```

```
0
1
2
3
4
```

```
# Prints out only odd numbers - 1,3,5,7,9
for x in range(10): # 0, 1, 2, 3... 9
    # Check if x is even
    if x % 2 == 0: # 2 % 2 = 0.. False
        continue #go back to the loop
    print(x) #1
```

```
1
3
5
7
9
```

```python
# What if I only want to print orange?
fruits = ['orange', 'banana', 'apple', 'pineapple']
i = 0
while i < len(fruits):
    print(fruits[i])
    if i > 0: #what is wrong?
        break
    i += 1 # what happens if we comment this out?
```

```
orange
banana
```

```python
# What if I only want to print orange?
fruits = ['orange', 'banana', 'apple', 'pineapple']
i = 0
while i < len(fruits):
    print(fruits[i])
    i += 1
    if i > 0:
        break
```

```
orange
```

```python
# What if I only want to print pineapple?
fruits = ['orange', 'banana', 'apple', 'pineapple']
i = 0
while i < len(fruits):
    if i == len(fruits)  - 1:
        print(fruits[i])
    i += 1
```

```
pineapple
```

```python
# we can also loop over strings like lists
string = 'apple'
for x in string:
    print(x)
```

```
a
p
p
l
e
```

`Exercise` on Loops

```python
# 1. Loop through and print out all even numbers from the numbers list
in the same order they are received.
# Don't print any numbers that come after 237 in the sequence.
```

```python
numbers = [
    951, 402, 984, 651, 360, 69, 408, 319, 601, 485, 980, 507, 725,
547, 544,
    615, 83, 165, 141, 501, 263, 617, 865, 575, 219, 390, 984, 592,
236, 105, 942, 941,
    386, 462, 47, 418, 907, 344, 236, 375, 823, 566, 597, 978, 328,
615, 953, 345,
    399, 162, 758, 219, 918, 237, 412, 566, 826, 248, 866, 950, 626,
949, 687, 217,
    815, 67, 104, 58, 512, 24, 892, 894, 767, 553, 81, 379, 843, 831,
445, 742, 717,
    958, 609, 842, 451, 688, 753, 854, 685, 93, 857, 440, 380, 126,
721, 328, 753, 470,
    743, 527
]

# 2. write a for loop to reverse a given string

# 3. Palindrome is a word which is read the same when you read it
forward and also backwards.
# write a conditoin to check whether a given word is palindrome or not

#this time you are not allowed to use slicing, you need to use loop

list=[['Lion','Deer','Sheep'],['Cub','Fawn','Lamb']]
# 4.print out all the animals one by one.

word='Automaticity'
#5. print the number of vowels

#6. print the word without the vowels. it should give you "tmtcty"

# 7. what is the output of the code below?
for n in [1, 2, 3, 4, 5, 6, 7]:
    if n % 3 == 0:      # n = 7
        print('a')
    elif n % 4 == 0:
        print('b')
    else:
        print('c')

#8 write python code to find multiples of 3 in the list
l = [1 ,2 ,3, 5, 6, 9, 7, 12]

#write your code here

#9 write python code to find the sum of the numbers in the list
ours = [ 1, 2, 4, 6]
```

```
#10 write a code to copy one list items to another list
l1 = [1, 2, 3, 5]
l2 = []

numbers = [
    951, 402, 984, 651, 360, 69, 408, 319, 601, 485, 980, 507, 725,
547, 544,
    615, 83, 165, 141, 501, 263, 617, 865, 575, 219, 390, 984, 592,
236, 105, 942, 941,
    386, 462, 47, 418, 907, 344, 236, 375, 823, 566, 597, 978, 328,
615, 953, 345,
    399, 162, 758, 219, 918, 237, 412, 566, 826, 248, 866, 950, 626,
949, 687, 217,
    815, 67, 104, 58, 512, 24, 892, 894, 767, 553, 81, 379, 843, 831,
445, 742, 717,
    958, 609, 842, 451, 688, 753, 854, 685, 93, 857, 440, 380, 126,
721, 328, 753, 470,
    743, 527
]

# given the above list of numbers, find the index of the number 831 by
using for loop and while loop
```